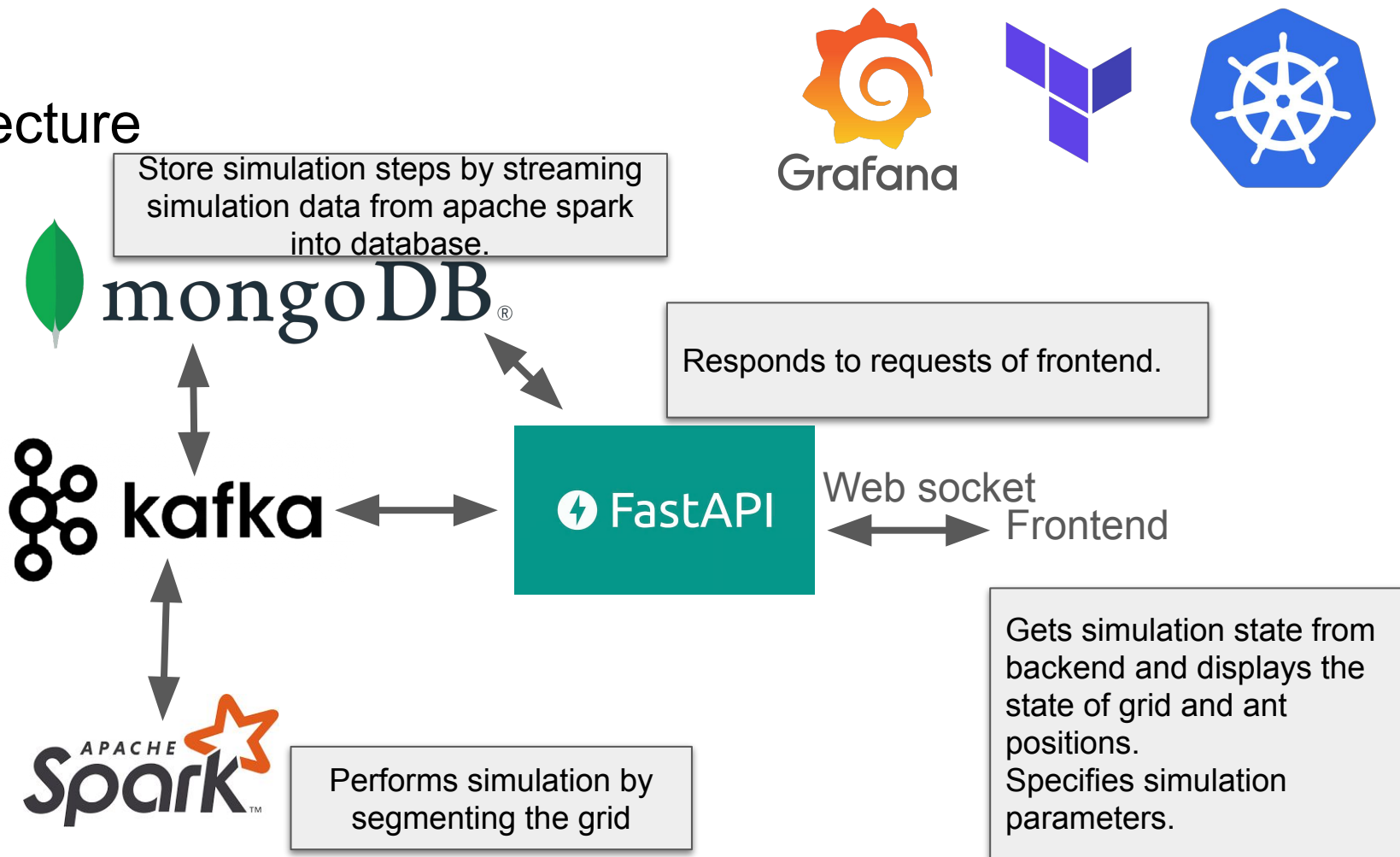


Requirements

- Initially, position ants, grid and set simulation parameters through an interface
- Simulate multiple ants according to [this paper](#)
- Application must be scalable, distributed, and fault tolerant
- Store the progression of the simulation (this can involve streaming)
- Display a history slice of the ants and grid in an interface
- Allow modification of ant position, grid and simulation parameters at a different point in time
- Rerun simulations starting from a point in time using the modified parameters.

Architecture



Real-Time Processing Flow

- User configures simulation → API sends task to Kafka
- Kafka sends job to Apache Spark for distributed execution
- Spark processes state changes & updates Redis
- Redis pushes updates to WebSockets → Frontend receives changes



Scalability & Fault Tolerance

- Kafka ensures event persistence (no lost updates)
- Spark auto-scales to handle increasing workloads
- Kubernetes manages deployment & scaling
- Redis minimizes latency for real-time updates